

When Django finds a URL that matches this pattern, it calls the view function `topic()` with the value assigned to `topic_id` as an argument. We'll use the value of `topic_id` to get the correct topic inside the function.

The Topic View

The `topic()` function needs to get the topic and all associated entries from the database, much like what we did earlier in the Django shell:

```
views.py  --snip--
❶ def topic(request, topic_id):
    """Show a single topic and all its entries."""
❷     topic = Topic.objects.get(id=topic_id)
❸     entries = topic.entry_set.order_by('-date_added')
❹     context = {'topic': topic, 'entries': entries}
❺     return render(request, 'learning_logs/topic.html', context)
```

This is the first view function that requires a parameter other than the request object. The function accepts the value captured by the expression `<int:topic_id>` and assigns it to `topic_id` ❶. Then we use `get()` to retrieve the topic, just as we did in the Django shell ❷. Next, we get all of the entries associated with this topic and order them according to `date_added` ❸. The minus sign in front of `date_added` sorts the results in reverse order, which will display the most recent entries first. We store the topic and entries in the context dictionary ❹ and call `render()` with the request object, the `topic.html` template, and the context dictionary ❺.

NOTE

The code phrases at ❷ and ❸ are called queries, because they query the database for specific information. When you're writing queries like these in your own projects, it's helpful to try them out in the Django shell first. You'll get much quicker feedback in the shell than you would by writing a view and template and then checking the results in a browser.

The Topic Template

The template needs to display the name of the topic and the entries. We also need to inform the user if no entries have been made yet for this topic.

```
topic.html  {% extends 'learning_logs/base.html' %}

            {% block content %}

❶     <p>Topic: {{ topic.text }}</p>

            <p>Entries:</p>
❷     <ul>
❸         {% for entry in entries %}
            <li>
❹             <p>{{ entry.date_added|date:'M d, Y H:i' }}</p>
❺             <p>{{ entry.text|linebreaks }}</p>
            </li>
❻         {% empty %}
```